

Data Analytics (DA)

Module -3

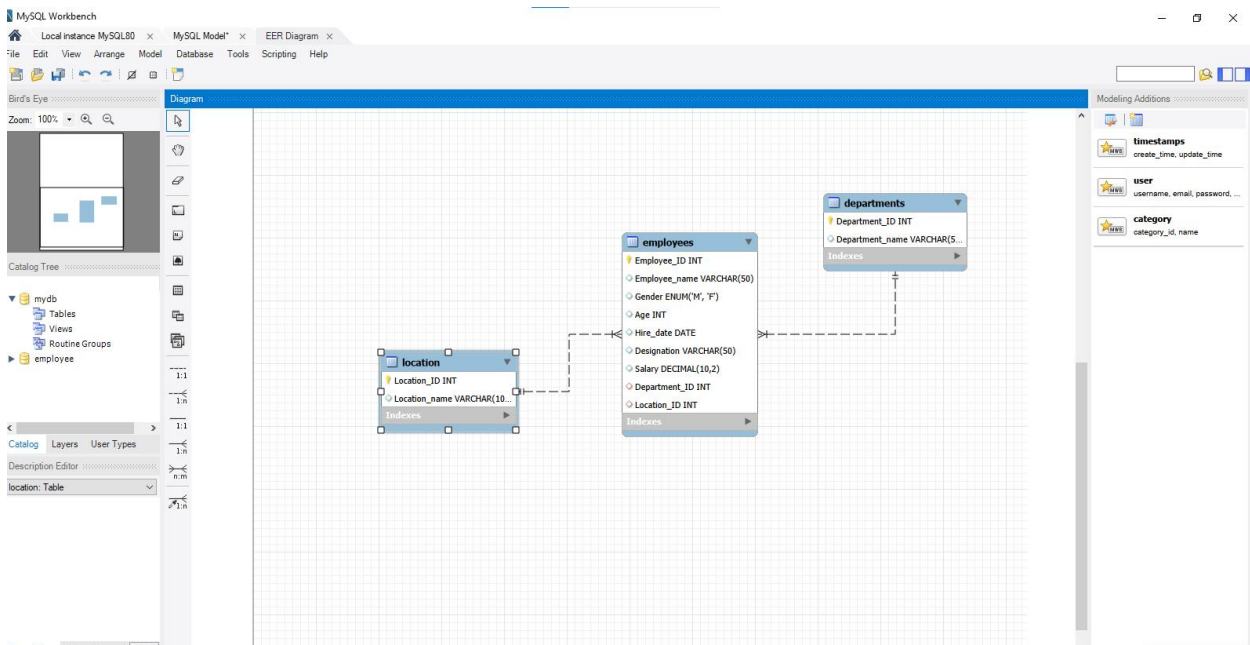
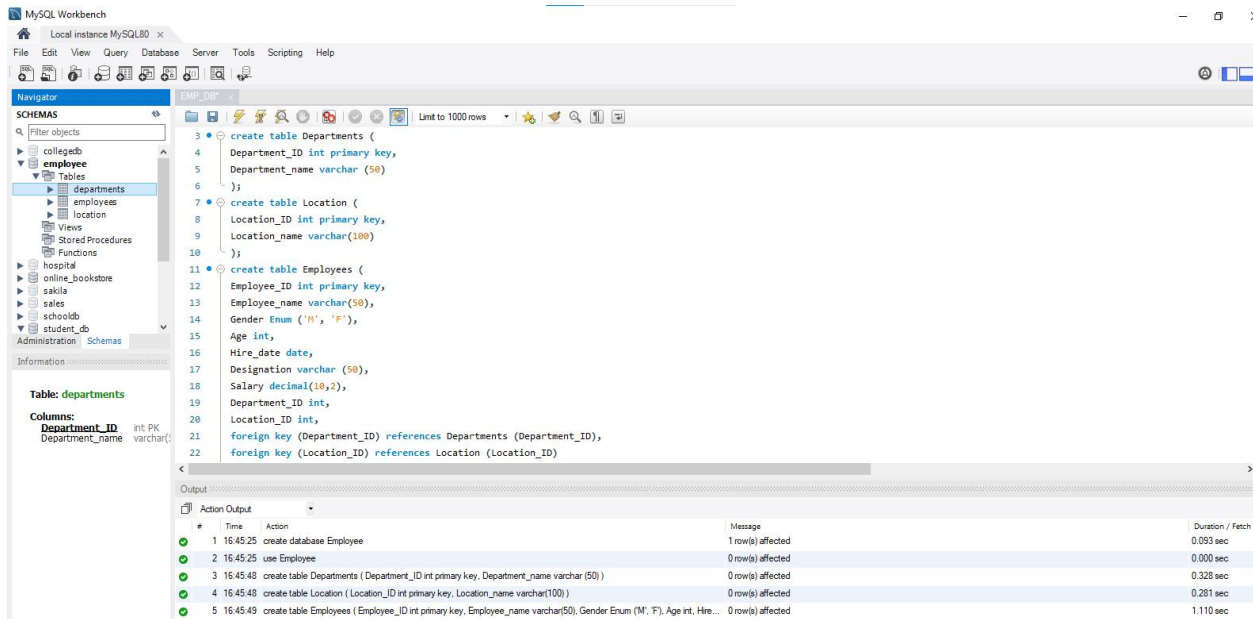
MySQL Assignment 1 -DDL Commands, Constraints

Tasks:

DDL Commands

1. Database and Table Creation (CREATE):

1. Create a Database for this project
2. Create all three tables in MySQL with appropriate data types and relationships.



2. Table Alteration (ALTER)

- Add a new column named "email" to the Employees table to store employee email addresses.
- Modify the data type of the "designation" column in the Employees table to support a wider range of values.
- Drop the "age" column from the Employees table.
- Rename the "hire_date" column to "date_of_joining".

3. Table Renaming (RENAME):

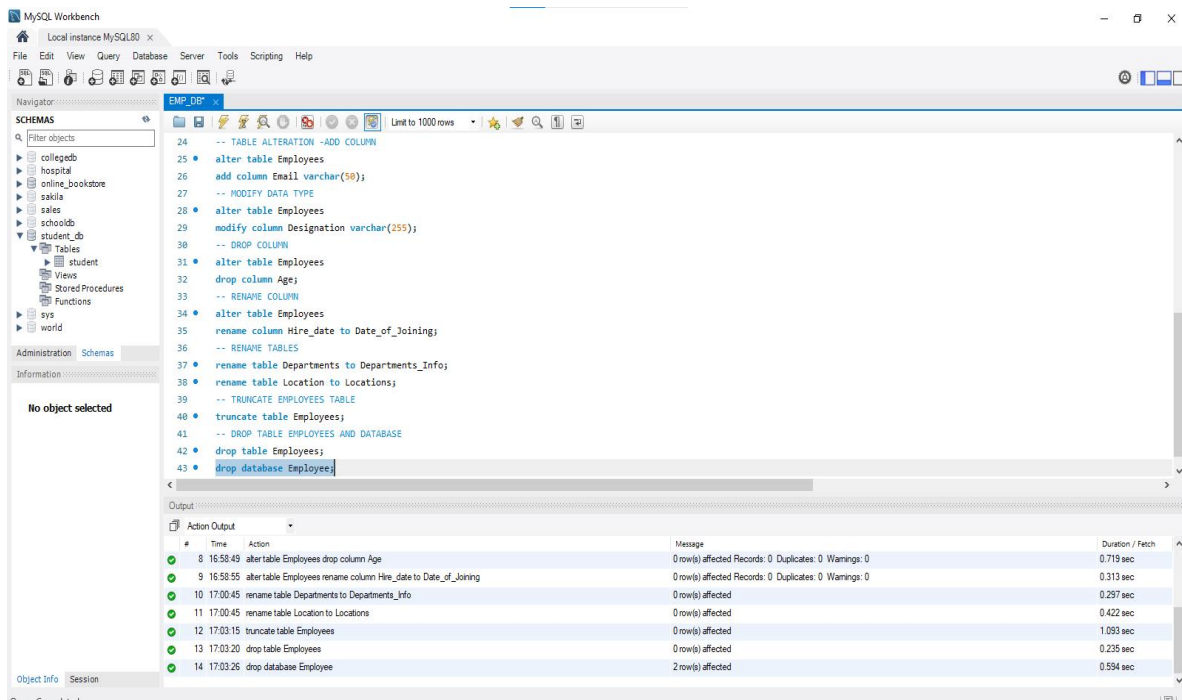
- Rename the "Departments" table to "Departments_Info".
- Rename the "Location" table to "Locations".

4. Table Truncation (TRUNCATE):

- Truncate the Employees table.

5. Database & Table Dropping (DROP):

- Drop the Employees table and then the "employee" database.



Tasks: Constraints

1. Database Recreation:

- Drop the 'employee' database if it exists
- Recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.
- Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.

2. Departments Table:

- Ensure that the "department_id" uniquely identifies each department.
- Set up constraints on the "department_name" to avoid duplicate and null entries.

3. Locations Table:

- Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.
- Implement constraints to prevent the insertion of null and duplicate locations.

4. Employees Table:

- Guarantee that each employee has a distinct identifier.
- Create a restriction to ensure that the employee's name is always provided.
- Limit the acceptable values for the "gender" field to only 'M' or 'F'.
- Enforce a condition to ensure that the employee's age is 18 or above.
- Automatically assign the current date to the "hire_date" field if not specified.

The screenshot shows the MySQL Workbench interface with the 'EMP_DB' database selected. The SQL editor contains the following commands:

```
1 drop database if exists Employee; -- Tasks: Constraints
2 create database Employee;
3 use Employee; -- RECREATE DEPARTMENTS, LOCATION, EMPLOYEES TABLE
4 create table Departments (
5     Departments_ID int primary key,
6     Departments_name varchar (50) NOT NULL UNIQUE
7 );
8 create table Locations (
9     Location_ID int AUTO_INCREMENT primary key,
10    Location_name varchar (30) NOT NULL UNIQUE
11 );
12 create table Employees (
13     Employee_ID int primary key,
14     Employee_name varchar(50) NOT NULL,
15     Gender Enum ('M', 'F') NOT NULL,
16     Age int check (Age >= 18),
17     Hire_date date DEFAULT (CURRENT_DATE),
18     Designation varchar (50),
19     Salary decimal(10,2),
20     Departments_ID int,
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	17:37:42	drop database if exists Employee	0 row(s) affected. 1 warning(s): 1008 Can't drop database 'employee': database doesn't exist	0.063 sec
2	17:38:14	create database Employee	1 row(s) affected	0.141 sec
3	17:38:14	use Employee	0 row(s) affected	0.000 sec
4	17:38:20	create table Departments (Departments_ID int primary key, Departments_name varchar (50) NOT NULL UNIQUE)	0 row(s) affected	0.469 sec
5	17:38:25	create table Locations (Location_ID int AUTO_INCREMENT primary key, Location_name varchar (30) NOT NULL UNIQUE)	0 row(s) affected	0.500 sec
6	17:38:33	create table Employees (Employee_ID int primary key, Employee_name varchar(50) NOT NULL, Gender Enum ('M', 'F') ...)	0 row(s) affected	0.500 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

collegedb

employee

Tables

Views

Stored Procedures

Functions

hospital

online_bookstore

sakila

sales

schooldb

student_db

Tables

student

Views

Administration Schemas

Information

No object selected

EMP_DB SQL File KEY DML LIVE

Limit to 1000 rows

```

12 create table Employees (
13     Employee_ID int primary key,
14     Employee_name varchar(50) NOT NULL,
15     Gender Enum ('M', 'F') NOT NULL,
16     Age int check (Age >= 18),
17     Hire_date date DEFAULT (CURRENT_DATE),
18     Designation varchar (50),
19     Salary decimal(18,2),
20     Departments_ID int,
21     Location_ID int,
22
23     foreign key (Departments_ID)
24     references Departments (Departments_ID)
25     ON DELETE CASCADE
26     ON UPDATE CASCADE,
27     foreign key (Location_ID)
28     references Locations (Location_ID)
29     ON DELETE CASCADE
30     ON UPDATE CASCADE
31 );

```

Output

#	Time	Action	Message	Duration / Fetch
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2	17:38:14	create database Employee	1 row(s) affected	0.141 sec
3	17:38:14	use Employee	0 row(s) affected	0.000 sec
4	17:38:20	create table Departments (Departments_ID int primary key, Departments_name varchar (50) NOT NULL UNIQUE)	0 row(s) affected	0.459 sec
5	17:38:25	create table Locations (Location_ID int AUTO_INCREMENT primary key, Location_name varchar (30) NOT NULL UNI...	0 row(s) affected	0.500 sec
6	17:38:33	create table Employees (Employee_ID int primary key, Employee_name varchar(50) NOT NULL, Gender Enum ('M', 'F') ...	0 row(s) affected	0.500 sec

Object Info Session